JSON

JavaScript Object Notation

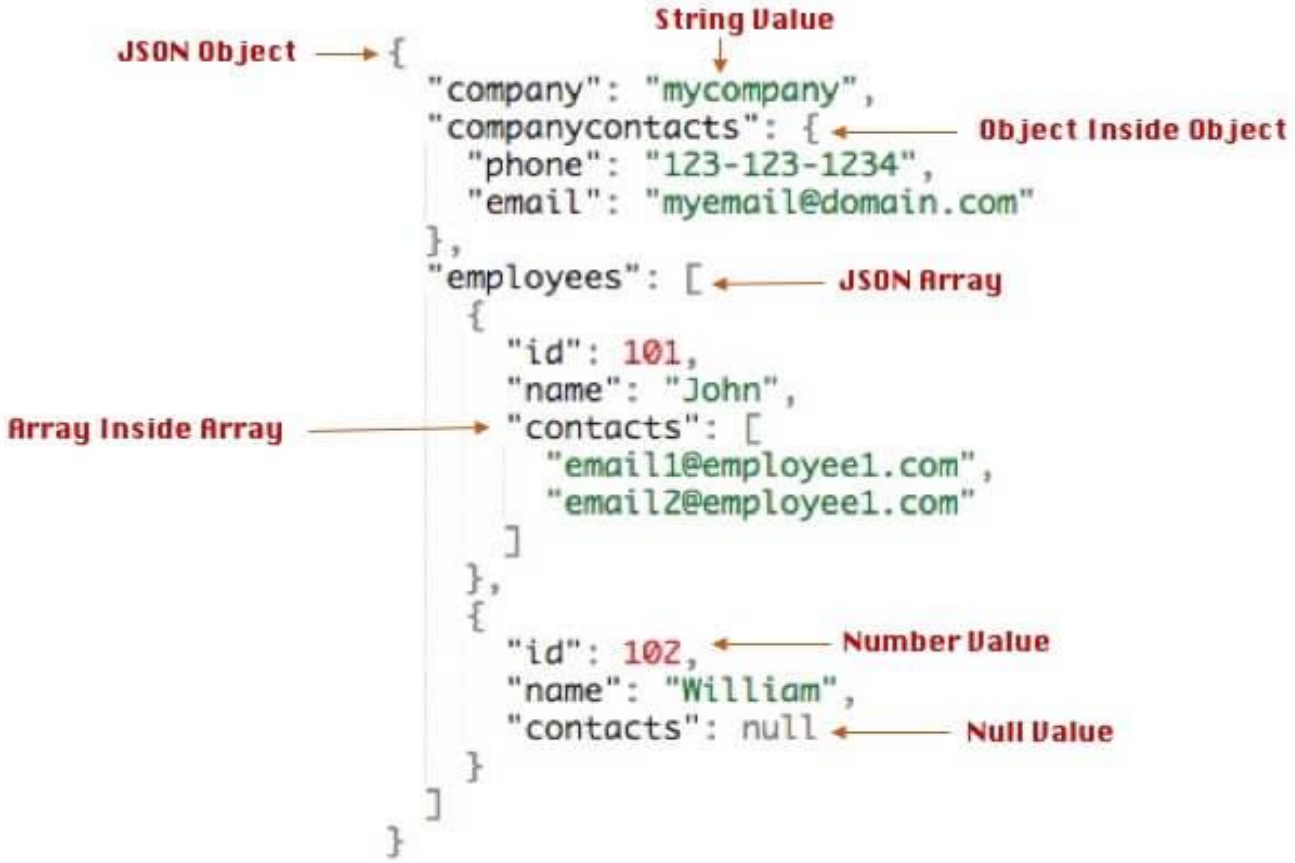
L J UNIVERSITY

JSON (JavaScript Object Notation)

- ✓ JSON stands for **J**ava**S**cript **O**bject **N**otation
- ✓ It is a **text format** for storing and transporting data
- ✓ It is "self-describing" and easy to understand
- ✓ It is a lightweight data-interchange format
- ✓ It is plain text written in JavaScript object notation
- ✓ It is language independent
- ✓ The JSON format is syntactically similar to the code for creating JavaScript objects.
- ✓ Because of this, a JavaScript program can easily convert JSON data into JavaScript objects.
- ✓ Since the format is text only, JSON data can easily be sent between computers, and used by any programming language.

JSON - Syntax

- ✓ Data is represented in key/value pairs.
- ✓ Curly braces hold objects and each name is followed by **':'(colon)**, the key/value pairs are separated by **, (comma).**
- ✓ Square brackets hold arrays and values are separated by **,(comma)**.



This example is a JSON string:

```
'{"name":"ABC", "age":20}'
```

It defines an JSON string with 2 properties: name and age

This example is a JSON object:

```
{
"name":"ABC",
"age":20
}

OR
{
name:"ABC",
age:20
}
```

It defines an JSON object with 2 properties: name and age

JSON Data Types

In JSON, values must be one of the following data types:

- a string
- a number
- an object (JSON object)
- an array
- a boolean
- *null*

JSON values cannot be one of the following data types:

- a function
- a date
- *undefined*

JSON Strings

Strings in JSON must be written in double quotes.

Example

```
{"name":"LJU"}
```

JSON Numbers

Numbers in JSON must be an integer or a floating point.

Example

```
{"age":30}
```

JSON Objects

Values in JSON can be objects.

Example

```
{
"employee":{"name":"LJU", "age":30, "city":"Ahmedabad"}
}
```

JSON Arrays

Values in JSON can be arrays.

Example

```
{
"employees":["Jenish", "Ankit", "Prit"]
}
```

JSON Booleans

Values in JSON can be true/false.

Example

{"sale":true}

JSON null

Values in JSON can be null.

Example

{"middlename":null}

Accessing values

Method	Syntax	Example Code	Result	Explanation
Dot Notation	object.key	person.name	Rahul	Access a value directly using the key with a dot.
Bracket Notation	object["key"]	person["age"]	22	Access value using square brackets. Useful when key is stored in a variable.
Nested Object Access	object.key.key	student.address.city	Ahmedabad	Used when one JSON object is inside another object.
Array Inside JSON	object.array[index]	person.skills[1]	CSS	Access elements from an array stored inside a JSON object.

Example JSON Object

```
let person = {
  name: "Rahul",
  age: 22,
  city: "Ahmedabad",
  skills: ["HTML", "CSS", "JavaScript"],
  address: {
```

```
    state: "Gujarat",
    country: "India"
  }
};
```

Accessing Values from above example

Code	Output	Explanation
person.name	Rahul	Access value using dot notation
person["age"]	22	Access value using bracket notation
person.skills[0]	HTML	Access first element of skills array
person.address.state	Gujarat	Access nested JSON object value

Run code differently

Running JSON Code in Different Ways

Method	File Name	Code Example	Output	Explanation
Using HTML File	j1.html	<script> var a={"name":"Test","age":20}; console.log(a.age); </script>	20	JSON object is created inside a script tag and accessed using dot notation.
Using JavaScript File	j1.js	console.log(a.age)	25	JSON object is stored in a JS file and executed using Node.js.
Using JSON File	j1.json + 1.js	const data=require("./j1.json")	Test	JSON file is imported into JavaScript and accessed using properties.

1. Using HTML File

File: J1.html

```
<script>
var a = {
  "name": "Test",
  "age": 20
}
console.log(a.age)
</script>
```

Output

```
20
```

2. Using JavaScript File

File: j1.js

```
var a = {
  "name": "Test",       // String
  "age": 25,            // Number
```

```
  "ispass": false,      // Boolean
  "address": {          // Nested Object
    "city": "Ahmedabad",
    "zip": "380015"
  },
  "subjects": ["Math", "Science"],  // Array
  "year": null          // Null
}

console.log(a);
console.log(a.age);
console.log(a["address"]);
console.log(a["address"]["city"]);
```

Run Command

```
node j1.js
```

Output

```
{
  name: 'Test',
  age: 25,
  ispass: false,
  address: { city: 'Ahmedabad', zip: '380015' },
  subjects: [ 'Math', 'Science' ],
  year: null
}
25
{ city: 'Ahmedabad', zip: '380015' }
Ahmedabad
```

3. Using JSON File

File: j1.json

```
{
  "name": "Test",
  "age": 20
}
```

File: 1.js

```
const data = require("./j1.json"); // Import JSON file
console.log(data.name);
```

Run Command

```
node 1.js
```

Output

```
Test
```

JavaScript built-in functions

Function	Purpose	Syntax	Example Result
JSON.parse()	Converts **JSON string → JavaScript object**	JSON.parse(string)	'{"var1":"LJ"}' → {var1:"LJ"}
JSON.stringify()	Converts **JavaScript object → JSON string**	JSON.stringify(object)	{var1:"LJ"} → '{"var1":"LJ"}'

1. JSON.parse()

- JavaScript has a built-in function to **convert JSON string into a JavaScript object**.
- When data is received from a **web server or API**, it is usually in **string format**.
- JSON.parse() converts that string into a **usable JavaScript object**.

Example Data Received from Server

```
'{"var1":"LJ","var2":"University"}'
```

Example

```
<script>
let obj = JSON.parse('{"var1":"LJ","var2":"University"}');
console.log(obj);
console.log(obj.var1 + " " + obj.var2);
</script>
```

Output

```
{var1: 'LJ', var2: 'University'}
LJ University
```

2. JSON.stringify()

- When sending data to a **web server**, the data must be in **string format**.
- JSON.stringify() converts a **JavaScript object into a JSON string**.

JavaScript Object

```
{var1: 'LJ', var2: 'University'}
```

Example

```
<script>
let obj = JSON.stringify({var1: 'LJ', var2: 'University'});
```

```
console.log(obj);
</script>
```

Output

```
{"var1":"LJ","var2":"University"}
```

JSON in API Usage

Function	Used For
JSON.parse(string)	Converts **API response (string)** → JavaScript object
JSON.stringify(object)	Converts **JavaScript object** → JSON string for API request

Updating Values of Existing Object Keys

1. Using Dot Notation

Method	Syntax	Example Code	Output	Explanation
Dot Notation	object.key = newValue	let person={name:"Rahul",age:22}; person.age=30; console.log(person);	{name:'Rahul', age:30}	Updates the value of an existing key using dot (.).
Bracket Notation	object["key"] = newValue	let person={name:"Rahul",city:"Ahmedabad"}; person["city"]="Surat"; console.log(person);	{name:'Rahul', city:'Surat'}	Updates value using square brackets. Useful when key name contains spaces or special characters.
Using Variable Key	object[variable] = newValue	let person={name:"Rahul",age:22}; let key="age"; person[key]=25; console.log(person);	{name:'Rahul', age:25}	Updates object value when key name is stored in a variable.

Example

```
let person = {
  name: "Rahul",
  age: 22,
  city: "Ahmedabad"
};

person.age = 30;   // update value
console.log(person);
```

Output

{ name: 'Rahul', age: 30, city: 'Ahmedabad' }

2. Using Bracket Notation

Syntax	Example	Output
object["key"] = newValue	person["city"] = "Surat"	Updates city to Surat

Example

```
let person = {
  name: "Rahul",
  age: 22,
  city: "Ahmedabad"
};
person["city"] = "Surat";   // update value
console.log(person);
```

Output

{ name: 'Rahul', age: 22, city: 'Surat' }

3. Updating Using Variable Key

```
let person = {
  name: "Rahul",
  age: 22
};

let key = "age";
person[key] = 25;

console.log(person);
```

Output

{ name: 'Rahul', age: 25 }

Examples

Below is a simple example of an array of objects

```
{
  "book": [
    {
      "id": "01",
      "language": "Java",
      "edition": "third",
      "author": "Herbert"
    },
    {
      "id": "07",
      "language": "C++",
      "edition": "second",
      "author": "E.Balagurusamy"
    }
  ]
}
```

Example:

```
var cars = [{
   color: 'gray',
   model: '1',
   nOfDoors: 4
   },
   {
   color: 'yellow',
   model: '2',
   nOfDoors: 4
}];
var jsonCars = JSON.stringify(cars);
console.log(jsonCars[0].model)
console.log(cars[0].model)
```

Output:

```
[{"color":"gray","model":"1","nOfDoors":4},{"color":"yellow","model":"2","nOfDoors":4}]
undefined
1
```

Write simple JSON object and fetch the values of each properties.

```
<script>
 const a = {
  'Name': 'tree',
  'course': 'Intro',
  'content': ['1', 'B', 'C']
 }
 console.log(a.Name);
 console.log(a.course);
 console.log(a.content);
 console.log(a.content[1]);
</script>
```

Output:

```
tree
Intro
(3) ['1', 'B', 'C']
0:'1'
1:'B'
2:'C'
B
```

Write code to fetch the "Give again" value from user object.

```
<script>
const user = {
      "name": ["ABC", "DEF", "GHI"],
      "age": "28",
      "course": ["FSD-1", "DE", "FSD-2"],
      "address": ["T1", "T2", { "t3": "Give again" }]
   }
   console.log(user.address[2].t3)
  // console.log(user['address'][2]['t3'])
</script>
```

Output

Give again

From JSON object fetch the values as asked.

```
<script>
 const a= {
   'Datastructures':
   [
      {
         'Name': 'tree',
         'course':'Intro',
         'content':['1','B','C']
      },
      {
         "Name": "tree1",
         "course":"Intro1",
         "content":["1","B","C","d"]
      }
   ],
   "xyz":
   {
      "Name":"Graphics",
      "Topic":["BFS","CDF","Sorting"],
   }
  }
  console.log(a.Datastructures[1].Name); //tree1
  console.log(a.Datastructures[0].Name); //tree
  console.log(a.xyz.Name); //Graphics
  console.log(a.xyz.Topic); //['BFS', 'CDF', 'Sorting']
  console.log(a.xyz.Topic[0]); //BFS
  console.log(a.Datastructures[1]); //{  "Name": "tree1",  "course":"Intro1",
"content":["1","B","C","d"]  }
  console.log(a.Name); //undefined
  console.log(a.xyz); //{Name: 'Graphics', Topic: Array(3)}
  </script>
```

Write one JSON string with date property (yyyy-mm-dd) and print date in India standard time.

```
<script>
   const obj=JSON.parse(`{"name":"xyz","age":"17","dob":"1997-03-14"}`);
   console.log(obj.dob);
   a = new Date(obj.dob);
   console.log(a);
</script>
```

Output

Fri Mar 14 1997 05:30:00 GMT+0530 (India Standard Time)

Write one JSON string with date property (yyyy-mm-dd) and print date in India standard time While clicking on button.

```
<html>
<script>
function call()
{
 obj=JSON.parse('{"Name":"ABC","Age":18,"City":"Ahmedabad","DOB":"1990-08-24"}');
  obj=new Date(obj.DOB);
 // document.write(obj.DOB);
  document.getElementById("demo").innerHTML=obj
}
</script>
<body>
  <P id="demo">  </P>
  <button onclick="call()"> ClickMe </button>
</body>
</html>
```

Output:

Thu Aug 23 1990 17:00:00 GMT-0700 (Pacific Daylight Time)

ClickMe

Consider below JSON object and fetch values using for loop.(for…in/for…of).

```
{
  "FSD": [
    {
      "Topic": "HTML",
      "course": "Beginer",
      "content": ["tags", "table", "form"],
    },
    {
      "Topic": "CSS",
      "course": "Beginer",
      "content": ["tags", "table", "form"]
    }
  ]
}
```

```
<script>
 const sub =
 {
  "FSD": [
    {
      "Topic": "HTML",
      "course": "Beginer",
      "content": ["tags", "table", "form"],
    },
    {
      "Topic": "CSS",
      "course": "Beginer",
      "content": ["tags", "table", "form"]
    }
  ]
 };
document.write("// Using for…in loop <br>")
// Using for…in loop
for (x in sub.FSD) {
  for (y in sub.FSD[x]) {
```

```
    document.write(sub.FSD[x][y] + "<br>")
   }
  }
document.write("// Using for…of loop <br>")
// Using for…of loop
 for(x1 of sub.FSD) {
    document.write(x1.Topic + "<br>")
    document.write(x1.course + "<br>")
    document.write(x1.content + "<br>")
    console.log(x)
 }
</script>
```

Output:

```
// Using for…in loop
HTML
Beginer
tags,table,form
CSS
Beginer
tags,table,form
// Using for…of loop
HTML
Beginer
tags,table,form
CSS
Beginer
tags,table,form
```

Write a JSON object which contains array of 3 objects. Each object contains 2 properties a name and an age. Now, sort an array values by age in descending order. Print name and age in terminal as per the sorted age.

```
const student =
   [
      {
```

```
        name: "NAS",
        age: 25
    },
    {
        name: "ABC",
        age: 30
    },
    {
        name: "XYZ",
        age: 33
    }
]
for (let x = 0; x < student.length; x++){
    for (let y = 0; y < student.length; y++){

        if (student[x].age > student[y].age) {
            var temp;
            temp = student[y];
            student[y] = student[x];
            student[x] = temp;
        }
    }
}
for (let i = 0; i < student.length; i++){
    console.log(student[i].name, student[i].age);
}
```

Output:

XYZ 33

ABC 30

NAS 25

OR

```
const student =
    [
        {
            name: "PQR",
            age: 25
        },
        {
```

```
        name: "ABC",
        age: 30
    },
    {
        name: "XYZ",
        age: 33
    }
  ]
const a=student.sort((a,b)=>b.age-a.age)

for (x of a){
  console.log(x.name +" "+x.age)
}
```

Output:

XYZ 33
ABC 30
PQR 25

Write a script to define two JSON objects named as "division1" and "division2" having an array to store name of students. These names should be sorted alphabetically and should be displayed in console.

```
var test = {
  "division1": {
    "name":["Z","B","H"]
  },
  "division2": {
    "name" :["Y","A","G"]
  }
}
const div1_data = test.division1.name;
const div2_data = test.division2.name;

var combine_data = div1_data.concat(div2_data).sort();
console.log(combine_data);
```

Output:

['A', 'B', 'G', 'H', 'Y', 'Z']

Write a function named "firstlast" that takes an array and returns an object with below conditions.

1) **The first element of an array must be an object's key**

2) The last element of an array must be a key's value.

Input = ['abc','def','ghi','jkl']

Output = { abc: 'jkl' }

```
function firstlast(a) {
   var temp = {};
   temp[a[0]] = a[a.length - 1];
   return temp;
}

var data = ['abc', 'def', 'ghi', 'jkl'];
console.log(firstlast(data));
```

Output:

{ abc: 'jkl' }

Explaination:

a**[0]** → Accesses the **first element** of the array and treats it as a **key**.

a**[a.length - 1]** → Accesses the **last element** of the array and assigns it as a **value**.

temp[a[0]] = a[a.length - 1]; → Dynamically assigns the key-value pair in the temp object.

Write a function 'FirstAndLast' that takes in an array, and returns an object with:

1) the first element of the array as the object's key, and

2) the last element of the array as that key's value.

(Example input: ['ABC', 'DEF', 'Employee', 'Manager'] output: ABC : 'Manager')

```
function FirstAndLast(arr) {
  let result = {};
  let first = arr[0];              // first element
  result[first] = arr[arr.length - 1];    // set key and value
  return result;
}
// Example
let input = ['ABC', 'DEF', 'Employee', 'Manager'];
console.log(FirstAndLast(input));
```

Write a JS to store an array of objects having height and name. Display highest height.

```
const person = [
      {
         name: "PQR",
         height: 5.2
```

```
    },
    {
        name: "ABC",
        height: 5.5
    },
    {
        name: "XYZ",
        height: 6.0
    }]
console.log(Math.max(...person.map(obj => obj.height)));
```

Output:

6

Explanation

- ✓ **person.map(obj => obj.height)**
- ✓ .map() creates a **new array** containing only the height values. **Result:** [5.2, 5.5, 6.0]
- ✓ **Math.max(...array)** : The **spread operator (...)** spreads the array into individual values.
- ✓ Math.max(5.2, 5.5, 6.0) returns **6.0** (the highest value.

OR

```
const person = [
    {
        name: "PQR",
        height: 5.2
    },
    {
        name: "ABC",
        height: 5.5
    },
    {
        name: "XYZ",
        height: 6.0
    }]
const max_height =person.sort((a,b)=> b.height-a.height)[0]
console.log(max_height.height)
```

Output:

6

Explanation

- ✓ .sort((a, b) => b.height - a.height) sorts the array in descending order based on the height property.
- ✓ **The first element ([0]) is the tallest person. It returns { name: "XYZ", height: 6.0 }**
- ✓ So, **max_height.height** extracts and prints height which is 6.0.

Write a code to fetch the values of keys and print a sentence as "Hi! We are students of semester 4 of CSE branch".

```
<script>
const a = {
   "A" : "LJU",
   "B" : ["CSE", "IT" , "CE"],
   "C" : [
      {
         "D" : "Hi",
         "E" : ['are', 4, {'F' : ['semester', 'We']} ]
      }
   ],
   "G" : {
      "H" : "students",
      "I" : ["of","!"]
      },
   "J" : [{"K" :"Python", "L":"branch"},"FSD-2"]
}
document.write(a.C[0].D, a.G.I[1] + " " + a.C[0].E[2].F[1] +" " + a.C[0].E[0] + " " +a.G.H + " "
+a.G.I[0] + " "+ a.C[0].E[2].F[0] + " "+ a.C[0].E[1] + " " +a.G.I[0] + " "+ a.B[0] + " "+ a.J[0].L)

</script>
```

Output:

Hi! We are students of semester 4 of CSE branch

Reference

Get JSON object values using for loop and make HTML table to display these values.

```
{ "FSD": [
    {
      "Topic": "HTML",
      "course": "Beginer",
      "content": ["tags", "table", "form"],
    },
    {
      "Topic": "CSS",
      "course": "Beginer",
      "content": ["tags", "table", "form"]
    }
  ] }
```

```
const sub ={
  "FSD": [
    {
      "Topic": "HTML",
      "course": "Beginer",
      "content": ["tags", "table", "form"],
    },
    {
      "Topic": "CSS",
      "course": "Beginer",
      "content": ["tags", "table", "form"]
    }
  ] };
 var temp = "<table border='2px solid red'>";
 temp += "<tr>";
 for (x in sub.FSD[0]) {
   temp += "<th>" + x + "</th>";
 }
 temp += "</tr>";
 for (x in sub.FSD) {
   temp += "<tr>";
   for (y in sub.FSD[x]) {
     temp += "<td>" + sub.FSD[x][y] + "</td>";
   }
   temp += "</tr>";
 }
```

```
temp += "</table>";
document.write(temp);
```

Topic	course	content
HTML	Beginer	tags,table,form
CSS	Beginer	tags,table,form